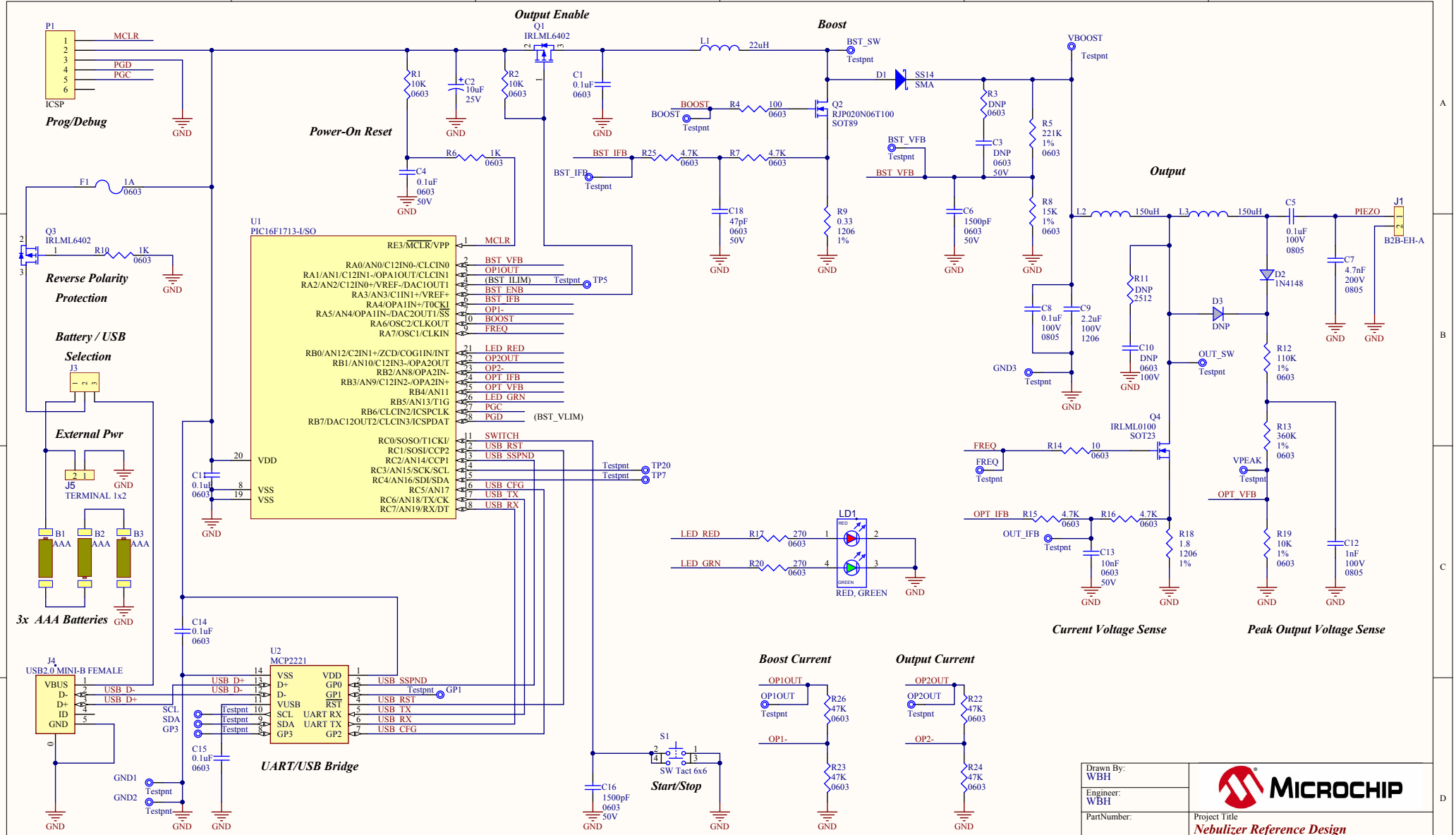
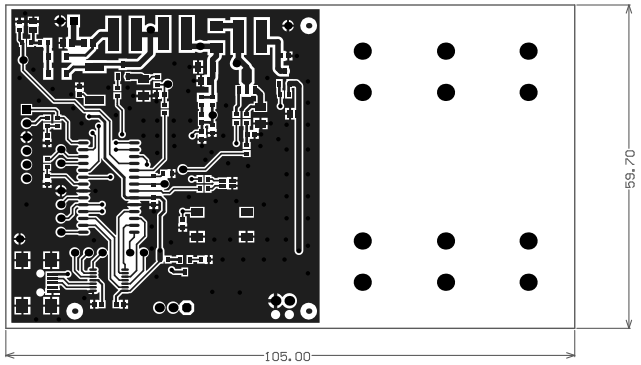

Prog/Debug
Power-On Reset
Output Enable
Boost
Output
Reverse Polarity Protection
Battery / USB Selection
External Pwr
3x AAA Batteries
UART/USB Bridge
Start/Stop
Op-amp Gain Networks
Current Voltage Sense
Peak Output Voltage Sense
Boost Current
Output Current
P1
ICSP
MCLR
PGD
PGC
Q1
IRLML6402
L1
22uH
BST_SW
VBOOST
D1
SS14
SMA
Q2
RJP020N06T100
SOT89
Q3
IRLML6402
Q4
IRLML0100
SOT23
U1
PIC16F1713-I/SO
U2
MCP2221
J4
USB2.0 MINI-B FEMALE
J5
TERMINAL 1x2
J1
B2B-EH-A
LD1
RED, GREEN
S1
SW Tact 6x6
PIEZO
OUT_SW
FREQ
VPEAK
OPT_VFB
OPT_IFB
OUT_IFB
BST_IFB
BST_VFB
BOOST
GND3
GND1
GND2
GND
Drawn By: WBH
Engineer: WBH
PartNumber:
Project Title
Nebulizer Reference Design
Sheet Title
Size B
Sch #:03-
Revision:B
Date: 2/27/2016 8:21:17 PM
Sheet 1 of 1
File: Nebulizer_MPG.SchDoc
Designed with Altium
Altium.com
MICROCHIP

REV	ECO	Comments	Date



THIS PCB TO BE MANUFACTURED TO MEET ALL ACCEPTANCE LEVELS OF A CLASS 2 PCB PER ANSI/IPC-A-600G.

MATERIAL: FR-4 or Equivalent

- [X] MULTILAYER [2] LAYERS [ ] CONTROLLED IMPEDANCE
- Cu WEIGHT EXTERNAL LAYERS [35um] FINISHED
- Cu WEIGHT INTERNAL LAYERS [N/A] FINISHED
- FINISHED OVERALL THICKNESS [1.6] mm ± [10] %
- COPPER THIEVING ALLOWED [ ] YES [X] NO

FINISH:

- [X] LEAD-FREE HOT AIR LEVELING
- [ ] IMMERSION GOLD
- [ ] IMMERSION TIN
- [ ] SMOBC WITH SELECTIVE GOLD PLATING ON LANDS INDICATED. 1um GOLD OVER 5-10 um NICKEL

SOLDERMASK [DYNACHEM EPIC 200 LPI OR EQUIVALENT]

SOLDERMASK COLOR [GREEN HIGH GLOSS]

SILKSCREEN COLOR [WHITE]

ALL HOLES TO BE LOCATED BY THE COORDINATES FROM THE NC DRILL DATA PROVIDED.

USE ARTWORK SET NO. [05-.ProjectBoardNumber] PCB REV [.ProjectRevisionART]

ALL UL LOGO, MANUFACTURER'S ID, AND DATE CODES SHALL BE PLACED ON THE BOTTOM SIDE UNLESS OTHERWISE INDICATED.

ANY ALTERNATIVES TO THE ABOVE SPECIFICATIONS MUST FIRST BE APPROVED.

Layer	Name	Material	Thickness	Constant	Board Layer Stack
1	Top Paste				
2	Top Overlay				
3	Top Solder	Solder Resist	0.010mm	3.5	
4	Top Layer	Copper	0.036mm		
5	Dielectric1	FR-4	1.500mm	4.8	
6	Bottom Layer	Copper	0.036mm		
7	Bottom Solder	Solder Resist	0.010mm	3.5	
8	Bottom Overlay				
9	Bottom Paste				

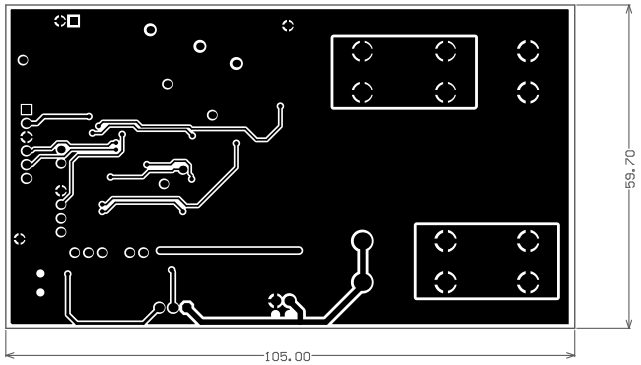
MICROCHIP
Microchip Technology Inc.
2355 W. Chandler Blvd.
Chandler, AZ 85224

TITLE: .ProjectTitle

PART NUMBER: .ProjectpartNumber

PCB DESIGNER:	GERBER FILE:		
.ProjectDrawnBy	Top Layer		
ENGINEER: .ProjectEngineer	BOARD NUMBER: 04-.ProjectBoardNumber		DATE: 2/27/2016
PCB FILE NAME: Nebulizer_MPGJ.PcbDoc	LAYER NAME: Top Layer (2)	REV: .ProjectRevisionART	10mm

REV	ECO	Comments	Date



THIS PCB TO BE MANUFACTURED TO MEET ALL ACCEPTANCE LEVELS OF A CLASS 2 PCB PER ANSI/IPC-A-600G.

MATERIAL: FR-4 or Equivalent

[X] MULTILAYER [2] LAYERS [ ] CONTROLLED IMPEDANCE

Cu WEIGHT EXTERNAL LAYERS [35um] FINISHED

Cu WEIGHT INTERNAL LAYERS [N/A] FINISHED

FINISHED OVERALL THICKNESS [1.6] mm ± [10] %

COPPER THIEVING ALLOWED [ ] YES [X] NO

FINISH:

[X] LEAD-FREE HOT AIR LEVELING

[ ] IMMERSION GOLD

[ ] IMMERSION TIN

[ ] SMOBC WITH SELECTIVE GOLD PLATING ON LANDS INDICATED. 1um GOLD OVER 5-10 um NICKEL

SOLDERMASK [DYNACHEM EPIC 200 LPI OR EQUIVALENT]

SOLDERMASK COLOR [GREEN HIGH GLOSS]

SILKSCREEN COLOR [WHITE]

ALL HOLES TO BE LOCATED BY THE COORDINATES FROM THE NC DRILL DATA PROVIDED.

USE ARTWORK SET NO. [05-.ProjectBoardNumber] PCB REV [.ProjectRevisionART]

ALL UL LOGO, MANUFACTURER'S ID, AND DATE CODES SHALL BE PLACED ON THE BOTTOM SIDE UNLESS OTHERWISE INDICATED.

ANY ALTERNATIVES TO THE ABOVE SPECIFICATIONS MUST FIRST BE APPROVED.

Layer	Name	Material	Thickness	Constant	Board Layer Stack
1	Top Paste				
2	Top Overlay				
3	Top Solder	Solder Resist	0.010mm	3.5	
4	Top Layer	Copper	0.036mm		
5	Dielectric1	FR-4	1.500mm	4.8	
6	Bottom Layer	Copper	0.036mm		
7	Bottom Solder	Solder Resist	0.010mm	3.5	
8	Bottom Overlay				
9	Bottom Paste				

.ProjectPCBNumber

MICROCHIP
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2355 W. Chandler Blvd.
Chandler, AZ 85224

TITLE: .ProjectTitle

PART NUMBER: .ProjectpartNumber

PCB DESIGNER: .ProjectDrawnBy

GERBER FILE: Bottom Layer

ENGINEER: .ProjectEngineer

BOARD NUMBER: 04-.ProjectBoardNumber

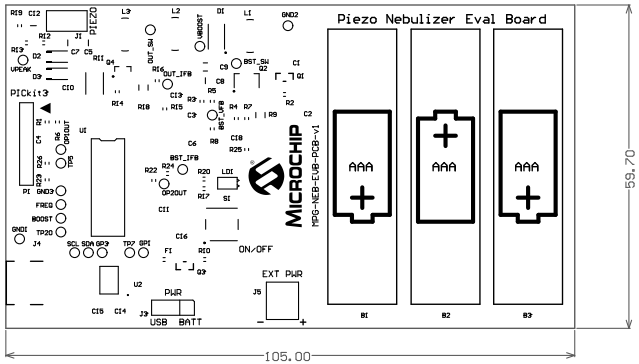
DATE: 2/27/2016

PCB FILE NAME: Nebulizer_MPG.PcbDoc

LAYER NAME: Bottom Layer

REV: .ProjectRevisionART

REV	ECO	Comments	Date



THIS PCB TO BE MANUFACTURED TO MEET ALL ACCEPTANCE LEVELS OF A CLASS 2 PCB PER ANSI/IPC-A-600G.

MATERIAL: FR-4 or Equivalent

- [x] MULTILAYER — 2 LAYERS — [ ] CONTROLLED IMPEDANCE
- Cu WEIGHT EXTERNAL LAYERS: 35um FINISHED
- Cu WEIGHT INTERNAL LAYERS: N/A FINISHED
- FINISHED OVERALL THICKNESS: 1.6 mm ± 10 %
- COPPER THIEVING ALLOWED: [ ] YES [X] NO

FINISH:

- [X] LEAD-FREE HOT AIR LEVELING
- [ ] IMMERSION GOLD
- [ ] IMMERSION TIN
- [ ] SMOBC WITH SELECTIVE GOLD PLATING ON LANDS INDICATED. 1um GOLD OVER 5-10 um NICKEL

SOLDERMASK: DYNACHEM EPIC 200 LPI OR EQUIVALENT

SOLDERMASK COLOR: GREEN HIGH GLOSS

SILKSCREEN COLOR: WHITE

ALL HOLES TO BE LOCATED BY THE COORDINATES FROM THE NC DRILL DATA PROVIDED.

USE ARTWORK SET NO. 05-.ProjectBoardNumber PCB REV .ProjectRevisionART

ALL UL LOGO, MANUFACTURER'S ID, AND DATE CODES SHALL BE PLACED ON THE BOTTOM SIDE UNLESS OTHERWISE INDICATED.

ANY ALTERNATIVES TO THE ABOVE SPECIFICATIONS MUST FIRST BE APPROVED.

Layer	Name	Material	Thickness	Constant	Board Layer Stack
1	Top Paste				
2	Top Overlay				
3	Top Solder	Solder Resist	0.010mm	3.5	
4	Top Layer	Copper	0.036mm		
5	Dielectric1	FR-4	1.500mm	4.8	
6	Bottom Layer	Copper	0.036mm		
7	Bottom Solder	Solder Resist	0.010mm	3.5	
8	Bottom Overlay				
9	Bottom Paste				

ICD

.ProjectPartNumber

MICROCHIP

dsPICDEM.net™ mTouch® chipKIT™
dsPICDEM™ MPLAB® dsPIC®
PICDEM.net™ PICkit™ PIC®
PICDEM™ PICtail™

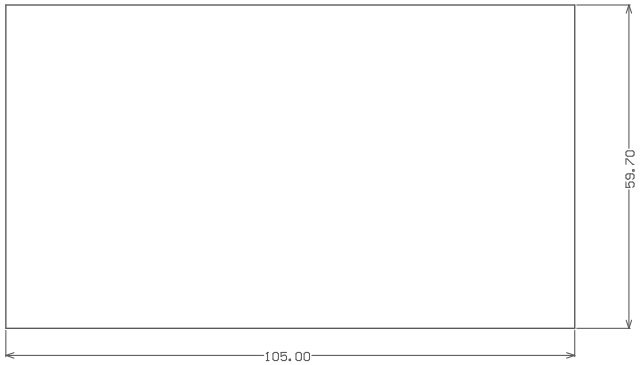
MICROCHIP
Microchip Technology Inc.
2355 W. Chandler Blvd.
Chandler, AZ 85224

TITLE: .ProjectTitle

PART NUMBER: .ProjectpartNumber

PCB DESIGNER: .ProjectDrawnBy	GERBER FILE: Top Silkscreen Overlay	
ENGINEER: .ProjectEngineer	BOARD NUMBER: 04-.ProjectBoardNumber	DATE: 2/27/2016
PCB FILE NAME: Nebulizer_MPG.PcbDoc	LAYER NAME: Top Overlay	REV: .ProjectRevisionART

REV	ECO	Comments	Date



THIS PCB TO BE MANUFACTURED TO MEET ALL ACCEPTANCE LEVELS OF A CLASS 2 PCB PER ANSI/IPC-A-600G.

MATERIAL: FR-4 or Equivalent

[X] MULTILAYER [2] LAYERS [ ] CONTROLLED IMPEDANCE

Cu WEIGHT EXTERNAL LAYERS [35um] FINISHED

Cu WEIGHT INTERNAL LAYERS [N/A] FINISHED

FINISHED OVERALL THICKNESS [1.6] mm ± [10] %

COPPER THIEVING ALLOWED [ ] YES [X] NO

FINISH:

[X] LEAD-FREE HOT AIR LEVELING

[ ] IMMERSION GOLD

[ ] IMMERSION TIN

[ ] SMOBC WITH SELECTIVE GOLD PLATING ON LANDS INDICATED. 1um GOLD OVER 5-10 um NICKEL

SOLDERMASK [DYNACHEM EPIC 200 LPI OR EQUIVALENT]

SOLDERMASK COLOR [GREEN HIGH GLOSS]

SILKSCREEN COLOR [WHITE]

ALL HOLES TO BE LOCATED BY THE COORDINATES FROM THE NC DRILL DATA PROVIDED.

USE ARTWORK SET NO. [05-.ProjectBoardNumber] PCB REV [.ProjectRevisionART]

ALL UL LOGO, MANUFACTURER'S ID, AND DATE CODES SHALL BE PLACED ON THE BOTTOM SIDE UNLESS OTHERWISE INDICATED.

ANY ALTERNATIVES TO THE ABOVE SPECIFICATIONS MUST FIRST BE APPROVED.

Layer	Name	Material	Thickness	Constant	Board Layer Stack
1	Top Paste				
2	Top Overlay				
3	Top Solder	Solder Resist	0.010mm	3.5	
4	Top Layer	Copper	0.036mm		
5	Dielectric1	FR-4	1.500mm	4.8	
6	Bottom Layer	Copper	0.036mm		
7	Bottom Solder	Solder Resist	0.010mm	3.5	
8	Bottom Overlay				
9	Bottom Paste				

MICROCHIP
Microchip Technology Inc.
2355 W. Chandler Blvd.
Chandler, AZ 85224

TITLE: .ProjectTitle

PART NUMBER: .ProjectpartNumber

PCB DESIGNER: .ProjectDrawnBy

GERBER FILE: Bottom Silkscreen Overlay

ENGINEER: .ProjectEngineer

BOARD NUMBER: 04-.ProjectBoardNumber

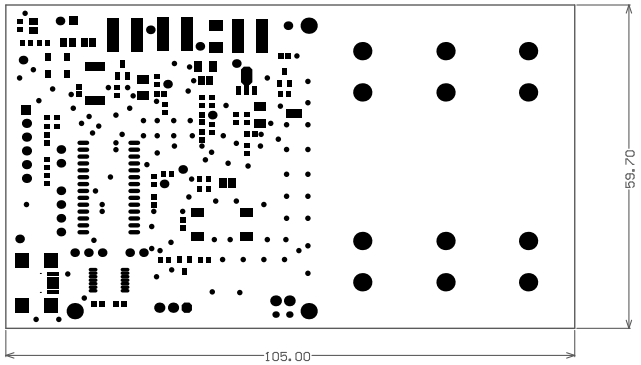
DATE: 2/27/2016

PCB FILE NAME: Nebulizer_MPG.PcbDoc

LAYER NAME: Bottom Overlay

REV: .ProjectRevisionART

REV	ECO	Comments	Date



THIS PCB TO BE MANUFACTURED TO MEET ALL ACCEPTANCE LEVELS OF A CLASS 2 PCB PER ANSI/IPC-A-600G.

MATERIAL: FR-4 or Equivalent

[X] MULTILAYER [2] LAYERS [ ] CONTROLLED IMPEDANCE

Cu WEIGHT EXTERNAL LAYERS [35um] FINISHED

Cu WEIGHT INTERNAL LAYERS [N/A] FINISHED

FINISHED OVERALL THICKNESS [1.6] mm ± [10] %

COPPER THIEVING ALLOWED [ ] YES [X] NO

FINISH:

[X] LEAD-FREE HOT AIR LEVELING

[ ] IMMERSION GOLD

[ ] IMMERSION TIN

[ ] SMOBC WITH SELECTIVE GOLD PLATING ON LANDS INDICATED. 1um GOLD OVER 5-10 um NICKEL

SOLDERMASK [DYNACHEM EPIC 200 LPI OR EQUIVALENT]

SOLDERMASK COLOR [GREEN HIGH GLOSS]

SILKSCREEN COLOR [WHITE]

ALL HOLES TO BE LOCATED BY THE COORDINATES FROM THE NC DRILL DATA PROVIDED.

USE ARTWORK SET NO. [05-.ProjectBoardNumber] PCB REV [.ProjectRevisionART]

ALL UL LOGO, MANUFACTURER'S ID, AND DATE CODES SHALL BE PLACED ON THE BOTTOM SIDE UNLESS OTHERWISE INDICATED.

ANY ALTERNATIVES TO THE ABOVE SPECIFICATIONS MUST FIRST BE APPROVED.

Layer	Name	Material	Thickness	Constant	Board Layer Stack
1	Top Paste				
2	Top Overlay				
3	Top Solder	Solder Resist	0.010mm	3.5	
4	Top Layer	Copper	0.036mm		
5	Dielectric1	FR-4	1.500mm	4.8	
6	Bottom Layer	Copper	0.036mm		
7	Bottom Solder	Solder Resist	0.010mm	3.5	
8	Bottom Overlay				
9	Bottom Paste				

MICROCHIP
Microchip Technology Inc.
2355 W. Chandler Blvd.
Chandler, AZ 85224

TITLE: .ProjectTitle

PART NUMBER: .ProjectpartNumber

PCB DESIGNER: .ProjectDrawnBy

GERBER FILE: Top Solder Mask Print

ENGINEER: .ProjectEngineer

BOARD NUMBER: 04-.ProjectBoardNumber

DATE: 2/27/2016

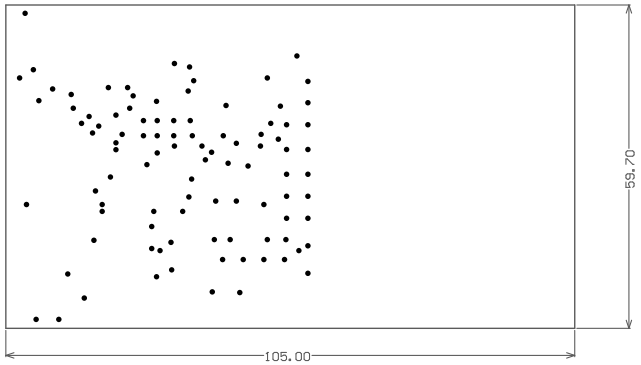
PCB FILE NAME: Nebulizer_MPG.PcbDoc

LAYER NAME: Top Solder (2)

REV: .ProjectRevisionART

10mm

REV	ECO	Comments	Date



THIS PCB TO BE MANUFACTURED TO MEET ALL ACCEPTANCE LEVELS OF A CLASS 2 PCB PER ANSI/IPC-A-600G.

MATERIAL: FR-4 or Equivalent

[X] MULTILAYER [2] LAYERS [ ] CONTROLLED IMPEDANCE

Cu WEIGHT EXTERNAL LAYERS [35um] FINISHED

Cu WEIGHT INTERNAL LAYERS [N/A] FINISHED

FINISHED OVERALL THICKNESS [1.6] mm ± [10] %

COPPER THIEVING ALLOWED [ ] YES [X] NO

FINISH:

[X] LEAD-FREE HOT AIR LEVELING

[ ] IMMERSION GOLD

[ ] IMMERSION TIN

[ ] SMOBC WITH SELECTIVE GOLD PLATING ON LANDS INDICATED. 1um GOLD OVER 5-10 um NICKEL

SOLDERMASK [DYNACHEM EPIC 200 LPI OR EQUIVALENT]

SOLDERMASK COLOR [GREEN HIGH GLOSS]

SILKSCREEN COLOR [WHITE]

ALL HOLES TO BE LOCATED BY THE COORDINATES FROM THE NC DRILL DATA PROVIDED.

USE ARTWORK SET NO. [05-.ProjectBoardNumber] PCB REV [.ProjectRevisionART]

ALL UL LOGO, MANUFACTURER'S ID, AND DATE CODES SHALL BE PLACED ON THE BOTTOM SIDE UNLESS OTHERWISE INDICATED.

ANY ALTERNATIVES TO THE ABOVE SPECIFICATIONS MUST FIRST BE APPROVED.

Layer	Name	Material	Thickness	Constant	Board Layer Stack
1	Top Paste				
2	Top Overlay				
3	Top Solder	Solder Resist	0.010mm	3.5	
4	Top Layer	Copper	0.036mm		
5	Dielectric1	FR-4	1.500mm	4.8	
6	Bottom Layer	Copper	0.036mm		
7	Bottom Solder	Solder Resist	0.010mm	3.5	
8	Bottom Overlay				
9	Bottom Paste				

MICROCHIP
Microchip Technology Inc.
2355 W. Chandler Blvd.
Chandler, AZ 85224

TITLE: .ProjectTitle

PART NUMBER: .ProjectpartNumber

PCB DESIGNER: .ProjectDrawnBy

GERBER FILE: Bottom Solder Mask Print

ENGINEER: .ProjectEngineer

BOARD NUMBER: 04-.ProjectBoardNumber

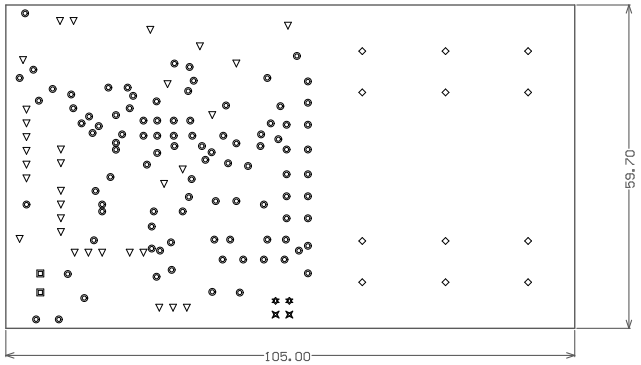
DATE: 2/27/2016

PCB FILE NAME: Nebulizer_MPG.PcbDoc

LAYER NAME: Bottom Solder

REV: .ProjectRevisionART

REV	ECO	Comments	Date



THIS PCB TO BE MANUFACTURED TO MEET ALL ACCEPTANCE LEVELS OF A CLASS 2 PCB PER ANSI/IPC-A-600G.

MATERIAL: FR-4 or Equivalent

- [X] MULTILAYER [2] LAYERS [ ] CONTROLLED IMPEDANCE
- Cu WEIGHT EXTERNAL LAYERS [35um] FINISHED
- Cu WEIGHT INTERNAL LAYERS [N/A] FINISHED
- FINISHED OVERALL THICKNESS [1.6] mm ± [10] %
- COPPER THIEVING ALLOWED [ ] YES [X] NO

FINISH:

- [X] LEAD-FREE HOT AIR LEVELING
- [ ] IMMERSION GOLD
- [ ] IMMERSION TIN
- [ ] SMOBC WITH SELECTIVE GOLD PLATING ON LANDS INDICATED. 1um GOLD OVER 5-10 um NICKEL

SOLDERMASK: DYNACHEM EPIC 200 LPI OR EQUIVALENT

SOLDERMASK COLOR: GREEN HIGH GLOSS

SILKSCREEN COLOR: WHITE

ALL HOLES TO BE LOCATED BY THE COORDINATES FROM THE NC DRILL DATA PROVIDED.

USE ARTWORK SET NO. 05-.ProjectBoardNumber PCB REV .ProjectRevisionART

ALL UL LOGO, MANUFACTURER'S ID, AND DATE CODES SHALL BE PLACED ON THE BOTTOM SIDE UNLESS OTHERWISE INDICATED.

ANY ALTERNATIVES TO THE ABOVE SPECIFICATIONS MUST FIRST BE APPROVED.

Symbol	Hit Count	Finished Hole Size	Plated	Hole Type
◘	2	0.900mm (35.43mil)	NPTH	Round
✖	2	1.100mm (43.31mil)	NPTH	Round
✡	2	1.100mm (43.31mil)	PTH	Round
◇	12	1.854mm (73.00mil)	PTH	Round
▽	32	0.900mm (35.43mil)	PTH	Round
◎	98	0.400mm (15.75mil)	PTH	Round
	148 Total			

Layer	Name	Material	Thickness	Constant	Board Layer Stack
1	Top Paste				
2	Top Overlay				
3	Top Solder	Solder Resist	0.010mm	3.5	
4	Top Layer	Copper	0.036mm		
5	Dielectric1	FR-4	1.500mm	4.8	
6	Bottom Layer	Copper	0.036mm		
7	Bottom Solder	Solder Resist	0.010mm	3.5	
8	Bottom Overlay				
9	Bottom Paste				

MICROCHIP
Microchip Technology Inc.
2355 W. Chandler Blvd.
Chandler, AZ 85224

TITLE: .ProjectTitle

PART NUMBER: .ProjectpartNumber

PCB DESIGNER: .ProjectDrawnBy

GERBER FILE: Drill Drawing For (Bottom Layer,Top Layer)

ENGINEER: .ProjectEngineer

BOARD NUMBER: 04-.ProjectBoardNumber

DOCUMENT NUMBER: .ProjectBoardNumber

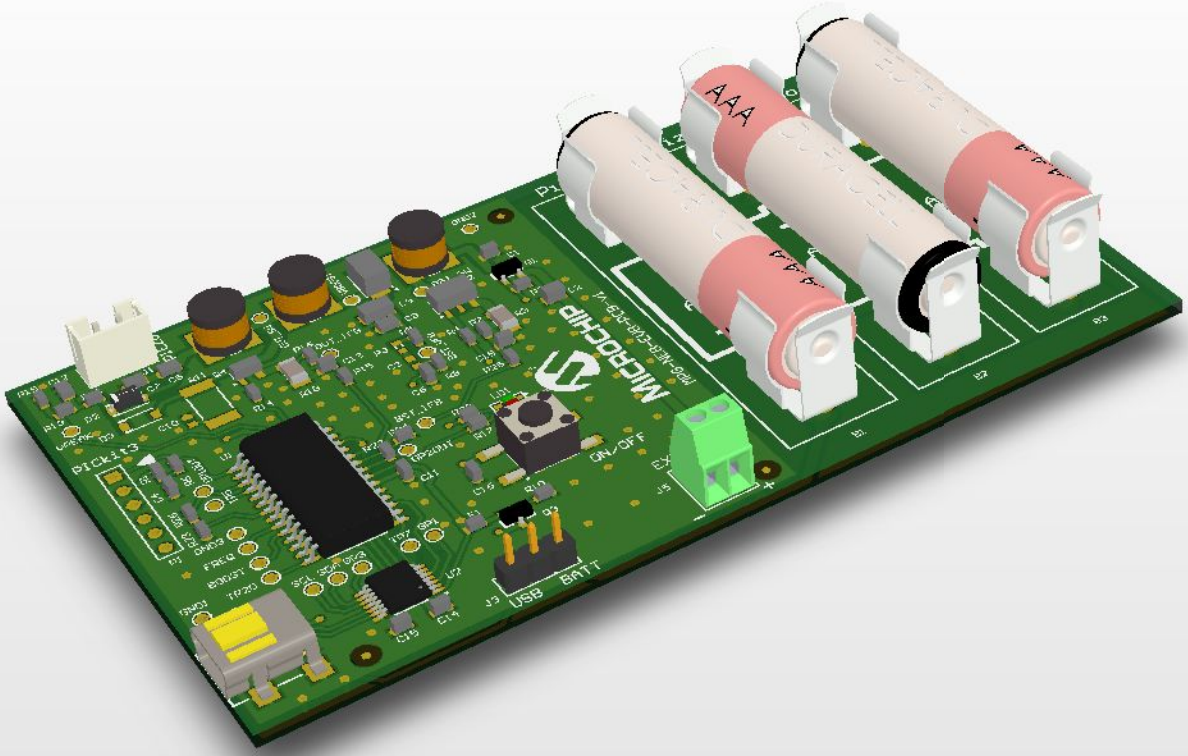
DATE: 2026-04-16

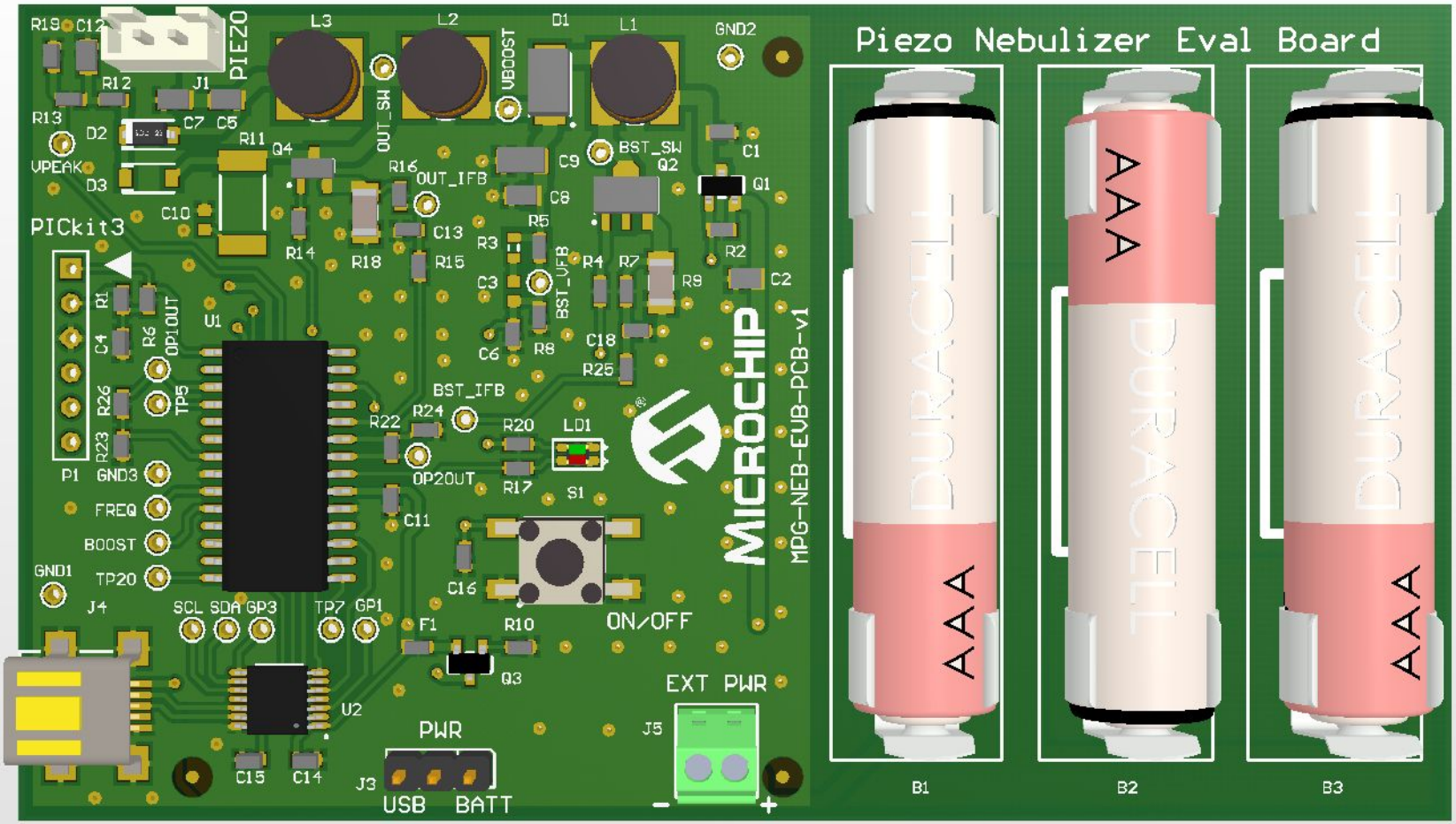
PCB FILE NAME: Nebulizer_MPG.PcbDoc

LAYER NAME: BOARD (M2)

REV: .ProjectRevisionFAB

10mm


AAA
MICROCHIP
ON/OFF
USB
BATT


Piezo Nebulizer Eval Board
MPG-NEB-EVB-PCB-v1
MICROCHIP
PIEZO
L3
L2
L1
D1
GND2
VBOOST
OUT_SW
BST_SW
OUT_IFB
BST_VFB
BST_IFB
OP1OUT
OP2OUT
VPEAK
PICkit3
P1
GND3
FREQ
BOOST
TP20
GND1
J4
SCL
SDA
GP3
TP7
GP1
U1
U2
LD1
S1
ON/OFF
EXT PWR
J5
PWR
J3
USB
BATT
B1
B2
B3
AAA
DURACELL